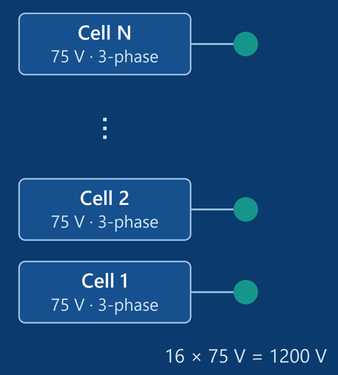
Project meeting · KTH Electric Power and Energy Systems · October 2026

Stacked Polyphase Bridge Traction Drives: from System Review to a GaN Building Block

Enes Ayaz, Shahriar Sarmast Ghahfarokhi, Kevin Klarin, Staffan Norrga

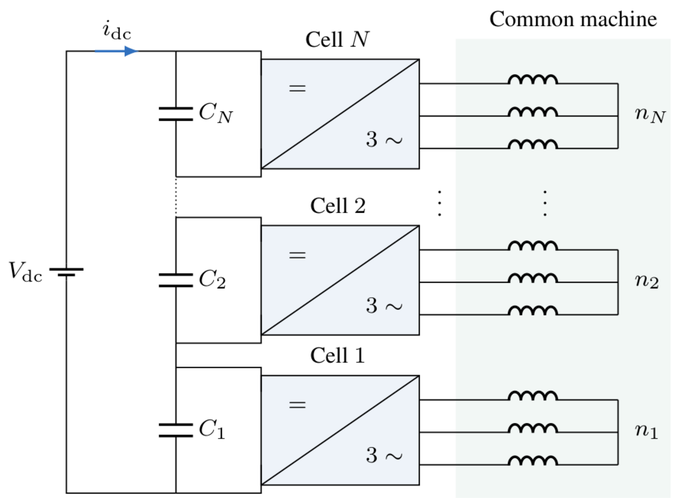
Division of Electric Power and Energy Systems, KTH Royal Institute of Technology, Stockholm, Sweden



1200 V	16 × 75 V	4 × EPC2361	0.41 nH	99.71 %	2.85 %
dc link, 300 kW traction	series cells, own winding groups	100 V GaN per switch	commutation loop per cell	semiconductor efficiency	shared-bus dc ripple, 50 kHz

A · Review: what the SPB architecture demands

1 Concept: split the dc voltage, keep one machine



N-cell SPB: series dc links, separate winding groups

2013–2014 **Topology and control** [37, 43]

2015–2017 **Stability and architectural assessment** [45, 46, 49]

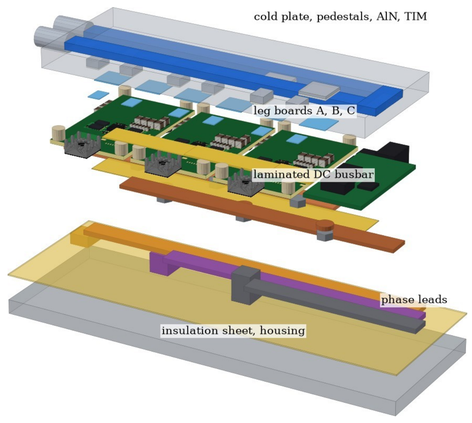
2017–2018 **Modulation and distributed control** [47, 48]

2019–2021 **Converter–machine coordination** [34, 50–52]

2023–Present **Reconfiguration and device integration** [53–58]

Open research direction
Unified SPB design for EV applications

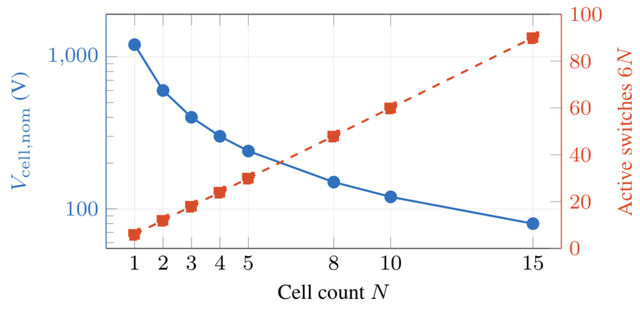
SPB research, 2013 to present



Integration concept: cold plate, legs, busbar

- N two-level bridges in series on the dc side; each feeds its own three-phase winding group.
- $V_{dc,k} \approx V_{dc}/N$: the device voltage class follows the cell, not the full link.
- All cells carry the same dc current; $P_{ac} = \Sigma P_k$, so cell voltages must be balanced actively.
- 6N switches, N drivers and capacitors: complexity moves to the system, it does not vanish.

2 Device voltage class and cell count

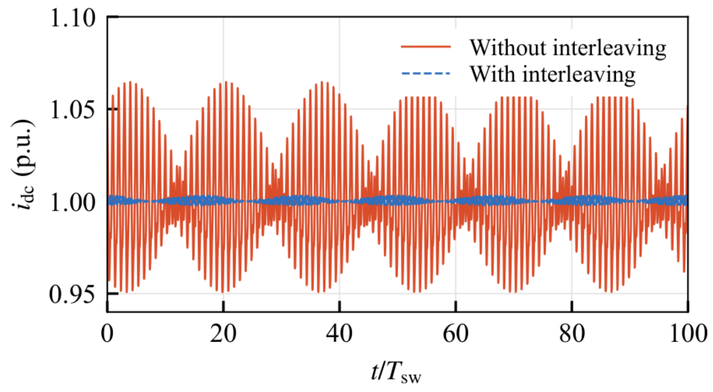


N	1	2	3	4	5	8	10	15
V_{cell} (V)	1200	600	400	300	240	150	120	80
$6N$	6	12	18	24	30	48	60	90

1200 V link: $V_{cell} = V_{dc}/N$ versus active switches 6N

- Lower blocking voltage raises the device figure of merit (D-FOM) within each technology.
- Rating with margin: $V_{rated} \geq k_{rel}(V_{dc,max}/N + \Delta V_{imb} + \Delta V_{sw})$.
- N = 16 gives 75 V cells for 100 V GaN; the overshoot L·di/dt sets the usable margin.

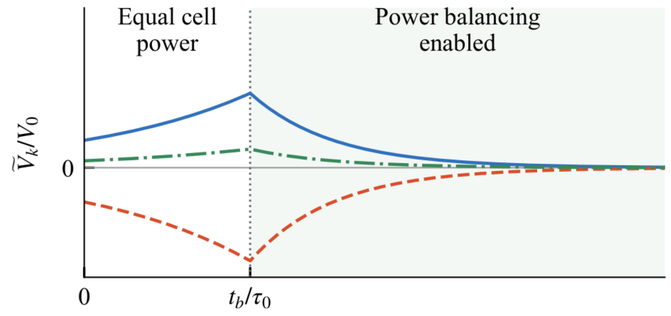
3 Modulation, ripple and common mode



Shared dc current of three cells, with and without interleaving

- Carrier interleaving $\delta_k = 2\pi(k-1)/N$ cancels carrier orders that are not multiples of N.
- Cell k sits at the offset $(k-1)V_{dc}/N$: a stack-position common-mode voltage modulation cannot remove.
- Complementary two-cell switching cancels the switching CMV only with matched timing.

4 Voltage balancing and fault handling

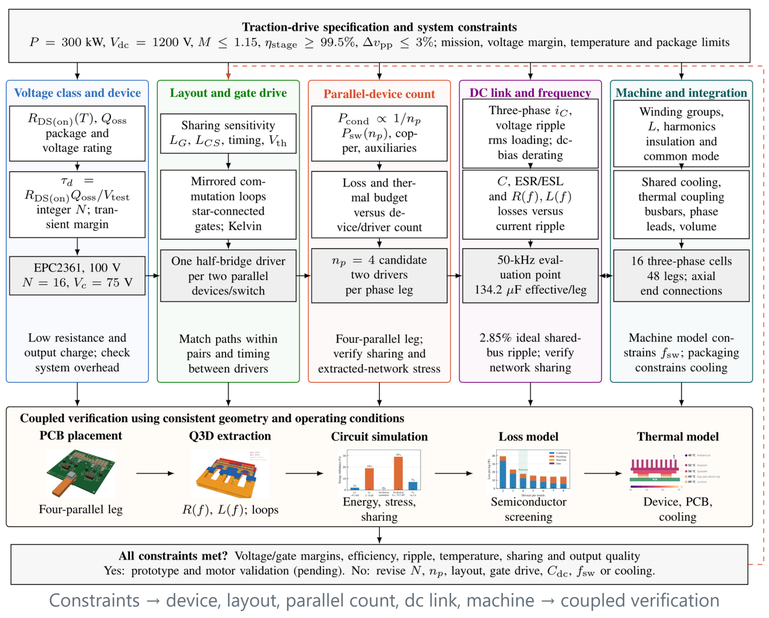


Differential power ΔP_k restores equal cell voltages (motoring)

- The open-loop series stack is unstable in motoring: N−1 positive real eigenvalues.
- Centralized, common-duty-ratio and distributed control trade authority against communication.
- A faulted cell must be bypassed without breaking the series dc path; survivors then carry V_{dc}/M.

B · Design flow and the four-parallel GaN building block

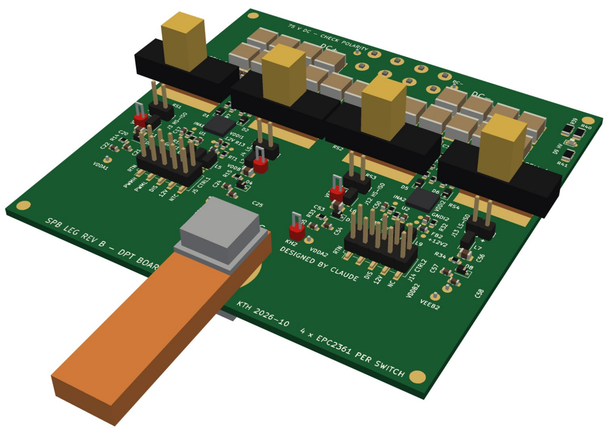
5 Iterative system design (review Fig. 16)



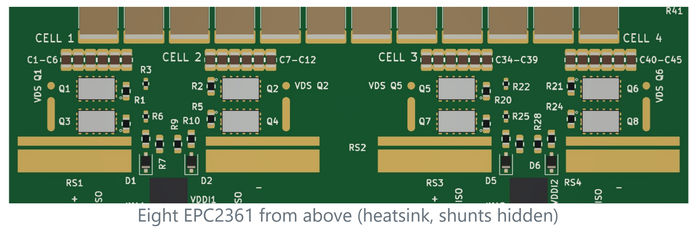
Constraints → device, layout, parallel count, dc link, machine → coupled verification

- This project: EPC2361, N = 16, n_p = 4 per switch, 50 kHz, 134 μF effective per leg; Q3D-extracted loops feed the LTspice double-pulse model.

6 Building block: final double-pulse-test board (Rev B)



DPT board: four cells, M6 ac bolt and copper bar



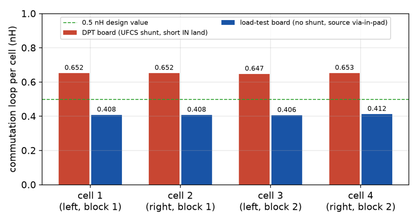
Eight EPC2361 from above (heatsink, shunts hidden)

- 89.5 × 85.2 mm, 4 layers, 2 oz, 0.1 mm L1–L2 dielectric: every loop closes vertically on the dc− plane.
- Four cells in two point-symmetric blocks; one 2EDF7275K per device pair, Kelvin gate returns.
- Per cell 20 × 22 μF 2220 + 10 × 1 μF; $R_{g,on}$ 3.5 Ω, −1.9 V off-state bias.
- 1 mΩ coaxial shunt per cell; MMCX and spring-probe pads for v_{DS}, v_{GS} of all eight devices.

LTspice with the extracted network (75 V, 322 A)

- High-side v_{DS} peak **92.9 V** < 100 V rating
- Four parallel high-side gate loops equal within 0.3 %
- Released for JLCPCB fabrication + assembly (DRC 0, final review 26 Oct)

7 Extracted parasitics (Ansys Q3D)



- Loop per cell: **0.41 nH** load board, 0.65 nH DPT (shunt); cells within 1.6 %
- Cell coupling k ≤ 0.034
- Gate loop, pair driven: 20.6 nH HS, 12.3 nH LS

8 Project status and next steps

- DPT board released for JLCPCB PCB + assembly; shunts and loop test points fitted in the lab.
- Double-pulse tests at 75 V up to 322 A: per-cell current sharing, overshoot, E_{on}/E_{off}, gate bump.
- Then three load-test legs + PCB busbar: one 75 V three-phase level, continuous and thermal tests.

9 Open research questions (review)

- Optimum N: device class versus component count shifts with SiC, GaN and cooling.
- Joint battery–converter–machine design on vehicle energy, mass, cost and reliability.
- Stack-position CMV, bearing currents and insulation across the full stack.

SPB moves the voltage problem from the device to the system: the building block shows that a 100 V GaN cell meets its loop and overshoot budget; the review sets out what the full stack must still prove — balancing, common mode, faults and machine co-design.

Enes Ayaz · KTH

[1] E. Ayaz et al., Stacked polyphase bridge converters for high-voltage traction drives: a review, IEEE OJPEL (submitted) [2] S. Norrga et al., IECON 2013 [3] EPC2361 datasheet [4] Rev B design-flow slides, 2026